

Technology Stack

Date	29 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Crop Recommendation System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, Bootstrap 5, JavaScript	Provides a responsive, user-friendly, and mobile-compatible interface with modern styling for easy interaction.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask offers a lightweight framework for integrating the machine learning model, handling requests, and exposing REST APIs efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset & Pickle Model Files	The crop recommendation dataset is stored in CSV format, while the trained Random Forest model is serialized using Pickle for fast loading and prediction.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Flask Server (Deployment-ready for Render/Heroku)	The application is developed and tested locally, with a modular architecture that can be easily deployed on cloud platforms in the future.
5	Version Control & CI/CD	Git & GitHub	Enables source code management, version tracking, team collaboration, and simplifies future CI/CD integration.
	<i>(e.g., GitHub, GitLab,</i>		

	<i>Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, Pandas, NumPy, Jupyter Notebook, Matplotlib, Seaborn	Used for data preprocessing, machine learning model training, evaluation, visualization, and exploratory data analysis.